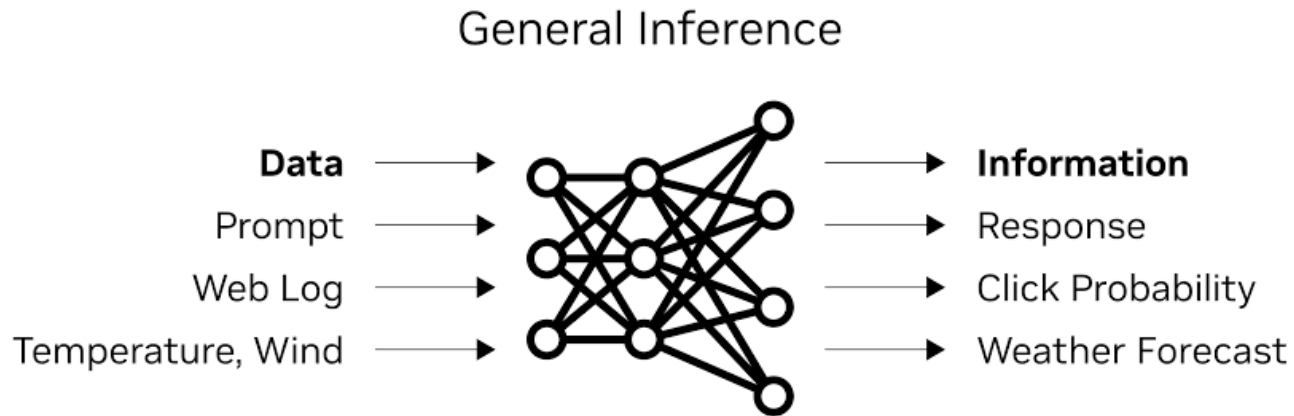


# AI Inference at Scale



## What is Inference?

- Using trained models to make predictions
- Convert data into actionable insights

## Cloud-Based Inference

- Perform thousands or millions of predictions simultaneously
- High-performance, scalable processing

## When is Cloud Inference Used?

- Fleet-wide analytics across devices
- Batch predictions & large datasets
- Global insights & trend analysis

## Cloud + Edge Synergy

- Edge inference → real-time decisions
- Cloud inference → complex & large-scale processing

**Scale intelligence across entire AIoT ecosystems**

# AI Inference at Scale

## Infrastructure Optimization

- ⚡ GPU clusters & specialized inference hardware
- ☁ Dedicated AI inference clouds & high-speed storage
- ✓ Deliver high throughput with low latency

## Model Optimization & Compression

- 🧠 Quantization & pruning reduce model size
- 🎯 Knowledge distillation preserves accuracy
- ✓ Faster execution & lower compute demand

## Efficient Serving Runtimes

- ⚙ High-performance inference engines
- 🔄 Continuous batching & GPU utilization
- ✓ Maximize throughput & performance

## Hybrid & Edge Architectures

- 📍 Deploy models closer to data sources
- ✓ Reduce latency & infrastructure costs

## Cost Optimization

- 💰 Optimize compute usage & scaling policies
- ✓ Control inference costs for production workloads

# Key Components in the Ecosystem



## Inference Scaling

- 📈 Increased inference calls improve responsiveness & performance
- ✓ Focus shifts from bigger models to smarter usage



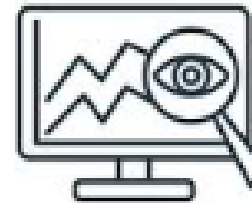
## Training-Inference Co-Optimization

- 🔄 Align training & inference workflows
- ✓ Improve efficiency & deployment performance



## Storage Infrastructure

- ⚡ High-IOPS, low-latency storage feeds GPUs efficiently
- ✓ Prevent data bottlenecks during inference



## Monitoring & Management

- 📊 Track performance & concurrency
- ⚙️ Manage models & deployment health
- ✓ Ensure reliability at scale

# Real-World Example: Large-Scale AI Inference



## Smart City Traffic Management

- 🚦 Millions of camera & sensor inputs analyzed in real time
- ⚡ AI detects congestion, violations & incidents
- ✓ Adjusts traffic signals dynamically
- ✓ Improves traffic flow & emergency response



## Fleet Management & Telematics

- 🚚 Vehicles stream location & engine data continuously
- 🤖 AI predicts breakdowns & unsafe driving patterns
- ✓ Optimizes routes & fuel efficiency



## Industrial Predictive Maintenance

- 🏭 Thousands of machines send vibration & temperature data
- ⚠️ AI predicts equipment failure across facilities
- ✓ Prevents downtime & production loss